STAT 5605 Homework 5

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1 Problem 1

Continuing Problem 4 from Homework 3, an analysis was performed to study the relationship between three explanatory variables, X_1, X_2, and X_3 and a response variable Y using the following model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon,$$

where $\epsilon \sim N(0, \sigma^2)$ independently for n observations. For this dataset, recall MSE $= 0.15306$ with 117 error degrees of freedom. For observation 10, we have the following SAS output:

Obs	Dep Var y	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Hat Diag H
10	3.1780	2.5816	0.0547	0.5964	0.387	1.540	0.0195

1.1 (i)

Compute $\hat{\sigma}^2_{(10)}$, an estimate of σ^2 based on the data with the 10th observation deleted. [Read Slide 15 of Note 6.]

We can compute $\hat{\sigma}^2_{(10)}$ externally by rearranging the expression given by Slide 15 in Note 6:

$$(n-k-1)\text{MSE} = (n-k-2)\text{MSE}_{(i)} + \frac{e_i^2}{1-h_{ii}},$$

To get an expression for $MSE_{(i)}$, which will estimate $\sigma^2_{(10)}$ here:

$$\text{MSE}_{(i)} = \hat{\sigma}^2_{(10)} = \frac{(n-k-1)\text{MSE} - \frac{e_i^2}{1-h_{ii}}}{(n-k-2)},$$

Now we can substitute in the provided values from the SAS output line and the other problem information. Given that the MLR model has x coefficients, and $n - p = 117$ error degrees of freedom, we can identify n (sample size) and k (number of non-intercept regression coefficients) as $n = 121$ and $k = p - 1 = 4 - 1 = 3$.

$$\text{MSE}_{(i)} = \hat{\sigma}^2_{(10)} = \frac{(121-3-1)(0.15306) - \frac{0.5964^2}{1-0.0195}}{(121-3-2)} = 0.151252,$$

1.2 (ii)

For observation 10, compute the *externally* Studentized residual t_{10}.

We can use the other formula from Slide 15 of Note 6 in order to calculate the externally Studentized residual for the tenth observation:

$$t_i = e_i \left[\frac{n-k-2}{\text{SSE}(1-h_{ii}) - e_i^2} \right]^{1/2}.$$

We know that we have nearly all the quantities we need from the prior subquestion, however, let's make sure that we have (or can find) the SSE. The SSE is not explicitly included in our given output, so we can quickly calculate it as $\text{SSE} = \text{MSE}(n-k-1) = 0.15306(121-3-1) = 17.90802$.
So, our externally Studentized 10th residual calculation becomes:

$$t_i = 0.5964 \left[\frac{121-3-2}{(17.90802)(1-0.0195) - 0.5964^2} \right]^{1/2} = 1.5487.$$

2 Problem 2

A substance used in biological and medical research is shipped by airflight to users in cartons of 1,000 ampules. The data below, involving 10 shipments, were collected on the number of times the carton was transferred from one aircraft to another over the shipment route (X) and the number of ampules found to be broken upon arrival (Y). Suppose that we fit a simple linear regression (SLR) model of Y on X to this data set.

i	1	2	3	4	5	6	7	8	9	10
X_i:	1	0	2	0	3	1	0	1	2	0
Y_i:	16	9	17	12	22	13	8	15	19	11

```
HW5PR2 <- read.table("./HW5PR2.txt", header=TRUE)
str(HW5PR2)
```

```
## 'data.frame':    10 obs. of  2 variables:
##  $ Y: num  16 9 17 12 22 13 8 15 19 11
##  $ X: num  1 0 2 0 3 1 0 1 2 0
```

```
n <- nrow(HW5PR2)
n
```

```
## [1] 10
```

```
PR2 <- lm(formula = Y ~ X, data=HW5PR2)
resid <- PR2$residuals
```

2.1 (a)

Plot the residual e_i against X_i to ascertain whether any departures from the SLR model are evident. What is your conclusion?

```
plot(HW5PR2$X, resid, pch = 20, main = "Residual vs. Predictor plot",
     xlab = "X", ylab = "residuals")
```

We plotted the residuals against the predictor in Figure 1. In the plot, we observe that the residuals have a funnel shape, narrowing as the plot moves to the right. This may indicate that we have an issue with our homoscedasticty assumption, suggesting that variance in the errors may decrease for increasing X. There are no very large residuals in this set, and the residuals appear to be centered around 0, although we may have fewer residuals close to zero than we might expect (this statement would have to be proven though). The plot does not give us reason to refute a linear relationship as we do not observe non-linear shapes in the residual plot. We should note that while there appears to be meaningful trends in the residuals in relation to the value of X, our sample size is very modest ($n = 10$) and patterns at this sample size may be spurious, underdeveloped, and/or misleading.

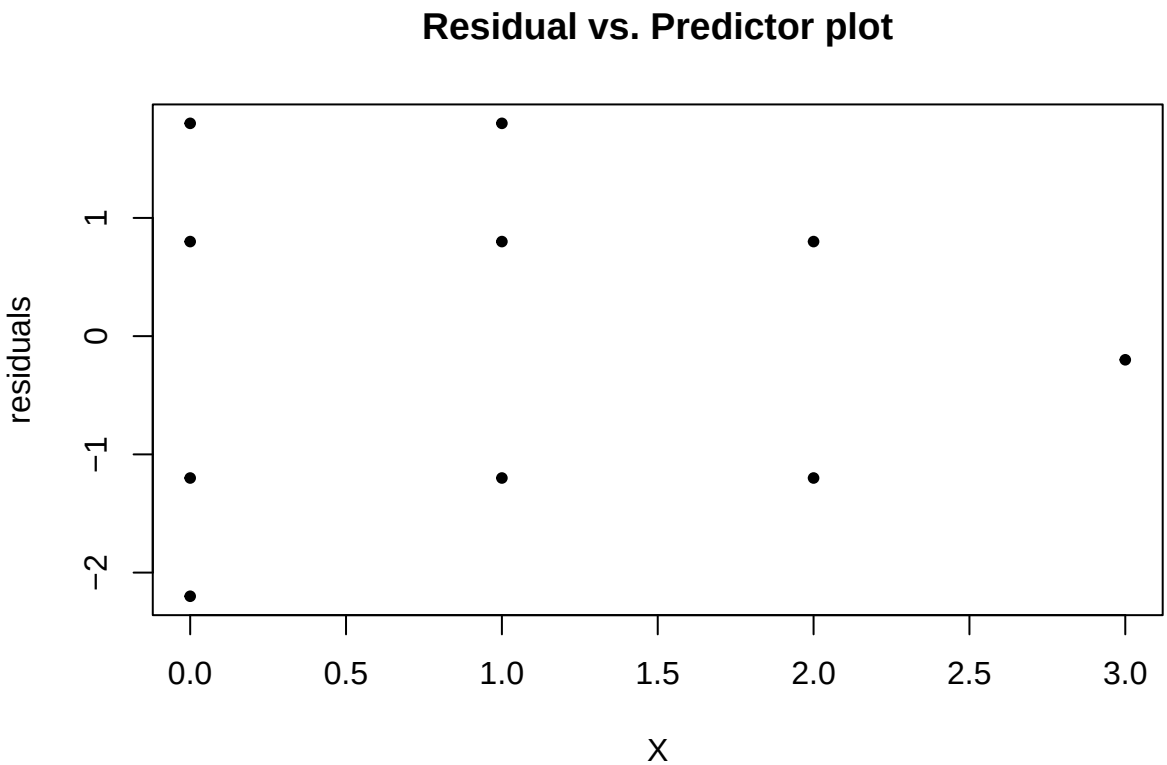


Figure 1: Question 2 Residuals versus Predictor Plot

2.2 (b)

Test the normality of the residuals using the Shapiro-Wilk Test and Kolmogorov-Smirnoff test, respectively. (Note that these tests are available in R; see Example 4 for relevant R code.) What do you conclude?

```r
# Shapiro-Wilk Test:
shapiro.test(resid)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid
## W = 0.90734, p-value = 0.2632
```

```r
# Kolmogorov-Smirnoff Test:
ks.test(resid, "pnorm")
```

```
## Warning in ks.test.default(resid, "pnorm"): ties should not be present for the
## one-sample Kolmogorov-Smirnov test
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
```

```
## data:  resid
## D = 0.28814, p-value = 0.3775
## alternative hypothesis: two-sided
```

Shapiro-Wilk Test
The Shapiro-Wilk Test is used to test the normality of residuals coming from independent observations after model fitting. It is most often used for small sample sizes.
We will assume that our observations for residuals are independent (this is not necessarily true as *errors should be independent*, but residuals are not necessarily independent). While the Shapiro-Wilk test is best used for continuous data, we will attempt to apply it to our problem. We will pre-specify a significance level of $\alpha = 0.05$ and use the following hypotheses:

$$H_0 : \text{The sample comes from a normally-distributed population, and}$$

$$H_1 : \text{The sample does not come from a normally-distributed population.}$$

We find a p-value of 0.2632 attained by our test. This means that we do not have evidence reject the null hypothesis which stated that the sample comes from a normally-distributed population. So, from this test we will not refute our normality assumption.
Kolmogorov-Smirnoff Test
The Kolmogorov-Smirnoff Test is another test for normality, and it is a nonparametric test which compares the empirical CDF of data from a sample to a CDF from a hypothesized distribution. The K-S test is typically used for moderate-to-large sample sizes. Our sample size of $n = 10$ is decidedly more small than we would desire for using this test. We can try to run the test, making assumptions of independence of observations again. Like the Shapiro-Wilk test, it is best to have continuous response data.
Let's pre-specify a significance level of $\alpha = 0.05$, and run the test under the hypotheses:

$$H_0 : \text{The residuals come from a N(0,1) distribution, and}$$

$$H_1 : \text{The residuals do not come from a N(0,1) distribution.}$$

Note that we specify a N(0,1) distribution in particular becuase the "ks.test(resid, "pnorm")" execution of the test assumes a N(0,1) distribution.
We find a p-value of 0.3775 for our test, which does not provide evidence to reject the null hypothesis. Therefore, we cannot refute that the residuals are standard normal distributed by the results of this question.
Note that while we failed to refute normality assumptions in this problem, it does not necessarily mean that the data is normal or that the residuals are truly N(0,1) distributed. Simply, we do not have the evidence to disprove these assumptions. Our tests were somewhat poorly employed, so we may be wary of their results.

2.3 (c)

In order to determine whether or not the error variance varies with the level of X, conduct the following tests. State your hypotheses and conclusion using the significance level of $\alpha = 0.05$.

2.3.1 (c1)

The Breusch-Pagan (BP) test assuming that $\log \sigma_i^2 = \gamma_0 + \gamma_1 X_i$ is applicable; and

```
# BP Test Implementation:
sse <- anova(PR2)[2, 2]
HW5PR2$resid <- PR2$residuals
HW5PR2$resid_sq <- HW5PR2$resid^2
diag_bp <- lm(resid_sq~X, data = HW5PR2)
# anova(diag_bp)
ssr_star <- anova(diag_bp)[1, 2]
chi_bp = (ssr_star/2)/(sse/n)^2
print(paste("test statistic:", chi_bp))
```

```
## [1] "test statistic: 1.03305785123967"
```

```
print(paste("critical chisq:", qchisq(0.95, 1)))
```

```
## [1] "critical chisq: 3.84145882069413"
```

```
print(paste("test statistic p-value:", pchisq(chi_bp, 1, lower.tail = FALSE)))
```

```
## [1] "test statistic p-value: 0.309441546906302"
```

The Breusch-Pagan (B-P) Test tests for heteroscedasticity in a linear regression model. Earlier, we were worried about heteroscedasticity when we looked at our residuals versus predictor values plot.
The B-P test relies on the assumption that errors are normally distributed and that observations are independent
Our hypotheses are:

$$H_0 : \text{Residuals observe homoscedasticity, and}$$

$$H_1 : \text{Residuals are heteroscedastic.}$$

Before running our test, we will decide upon a significance level threshold of $\alpha = 0.05$. Upon running our test, we find that we cannot reject the null hypothesis which stated that the residuals are homoscedastic (p-value $= 0.309$). Despite what we observed in our residual plot, this test does not provide evidence that an assumption of homoscedasticity is violated.

2.3.2 (c2)

The test of Chen, C.F. (1993) assuming $\text{Var}(\epsilon_i) = \sigma^2 \exp(\lambda X_i)$.

```
# C.F. Chen Test Implementation:
r_stand <- rstandard(PR2)
hii <- influence(PR2)$hat
HW5PR2$ui <- (10/8)*(1-hii)*r_stand^2
diag_ch <- lm(ui~X, data = HW5PR2)
s_chen <- anova(diag_ch)[1, 2]/2
print(paste("Test statistic:", s_chen))
```

```
## [1] "Test statistic: 1.03305785123967"
```

```
print(paste("Critical chi-squared value:", qchisq(0.95, 1)))
```

```
## [1] "Critical chi-squared value: 3.84145882069413"
```

```
print(paste("Corresponding p-value:", pchisq(s_chen, 1, lower.tail = FALSE)))
```

```
## [1] "Corresponding p-value: 0.309441546906302"
```

C.F. Chen's Test is another test for homoscedasticity. This test requires the assumption of independent, normally-distributed errors.
Our hypotheses for this test are the same as for the prior question:

$$H_0 : \text{Residuals observe homoscedasticity, and}$$

$$H_1 : \text{Residuals are heteroscedastic.}$$

Here, the results of the test are identical to the B-P test. Again we will fail to reject the null hypotheses.

3 Problem (3)

For the data, HW3PR1.txt, given in Problem 1 of Homework 3, answer the following questions.

```r
HW3PR1 <- read.table("./HW3PR1.txt", header=TRUE)
str(HW3PR1)
```

```
## 'data.frame':    50 obs. of  5 variables:
##  $ Y   : int  4264 4496 4317 4292 4945 4325 4110 4111 4161 4560 ...
##  $ X1  : int  305657 328476 317164 366745 265518 301995 269334 267631 296350 277223 .
##  $ X2  : num  7.17 6.2 4.61 7.02 8.61 6.88 7.23 6.27 6.49 6.37 ...
##  $ X3  : int  0 0 0 0 1 0 0 0 0 0 ...
##  $ case: int  1 2 3 4 5 6 7 8 9 10 ...
```

```r
nrow(HW3PR1)
```

```
## [1] 50
```

```r
q3_model <- lm(formula = Y ~ X1+X2+X3, data=HW3PR1)
summary(q3_model)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X3, data = HW3PR1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -267.84 -101.77  -22.09   80.93  297.79
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.100e+03  1.994e+02  20.563  < 2e-16 ***
## X1           9.343e-04  3.794e-04   2.463   0.0176 *
## X2          -1.200e+01  2.319e+01  -0.518   0.6073
## X3           6.189e+02  6.289e+01   9.842 6.73e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 143.2 on 46 degrees of freedom
## Multiple R-squared:  0.6985, Adjusted R-squared:  0.6788
## F-statistic: 35.52 on 3 and 46 DF,  p-value: 4.907e-12
```

```r
HW3b <- HW3PR1
HW3b$ti <- rstudent(q3_model)
# externally studentized residuals
b_ct <- qt(1-(0.05/50), 45)
b_ct
```

```
## [1] 3.28148
```

```r
HW3b[which(abs(HW3b$ti)>=b_ct),]
```

```
## [1] Y    X1   X2   X3   case ti
## <0 rows> (or 0-length row.names)
```

```r
HW3b$ei <- q3_model$residuals
HW3b$ri <- rstandard(q3_model)
HW3b$hii <- influence(q3_model)$hat
HW3b[which(HW3b$hii > 8/50),]
```

```
##       Y     X1   X2 X3 case         ti         ei         ri       hii
## 3  4317 317164 4.61  0    3 -0.1876591  -24.00543 -0.1896585 0.2190594
## 5  4945 265518 8.61  1    5  0.6332556   81.29671  0.6374192 0.2070615
## 16 4833 321773 5.82  1   16 -0.9438275 -116.73979 -0.9449496 0.2560147
## 21 4816 245674 7.72  1   21 -0.3071263  -39.84327 -0.3101952 0.1957635
## 22 4867 211944 6.45  1   22  0.2210247   27.43019  0.2233458 0.2647318
## 43 5045 369989 9.65  1   43  0.7944307   96.17197  0.7976353 0.2913495
## 44 4469 472476 8.20  0   44  0.2064978   25.96958  0.2086808 0.2450663
## 48 4993 442782 7.61  1   48 -0.3970381  -48.31580 -0.4007241 0.2913507
```

```r
durbinWatsonTest(q3_model, alternative = "positive")
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1      -0.1135048      2.172733   0.686
##  Alternative hypothesis: rho > 0
```

```r
dbetas <- dfbetas(q3_model)
HW3f <- HW3PR1
HW3f %>% select(case) %>% mutate(CooksD = cooks.distance(q3_model),dfits = dffits(q3_mo
bind_cols(HW3f, dbetas) -> HW3f
colnames(HW3f) <- c("case", "CooksD", "dfits","dbeta_X0", "dbeta_X1", "dbeta_X2", "dbeta
head(HW3f)
```

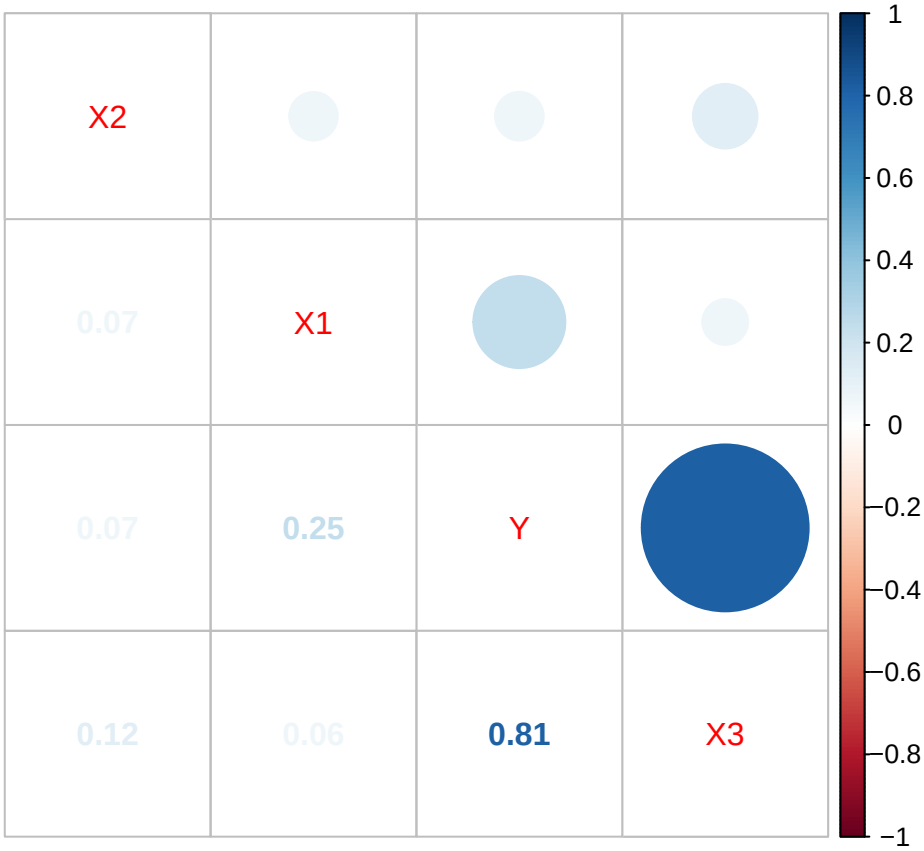
```
##   case       CooksD       dfits     dbeta_X0     dbeta_X1    dbeta_X2
## 1    1 0.0003806391 -0.03861996 -0.006764118 -0.004455918  0.00611068
## 2    2 0.0233119732  0.30669909  0.151297611  0.107512049 -0.22642765
## 3    3 0.0025224856 -0.09938974 -0.074118685 -0.015680404  0.09356948
## 4    4 0.0034960285 -0.11725315  0.018053680 -0.088324732  0.02858814
## 5    5 0.0265246813  0.32360032 -0.047773695 -0.089946518  0.11643136
## 6    6 0.0002310157  0.03007675  0.012670527  0.001971772 -0.01275268
##        dbeta_X3
## 1  0.0125471834
## 2 -0.0423736968
## 3  0.0005805273
## 4  0.0271649351
## 5  0.2612841427
```

```
## 6 -0.0079148549
```

```r
MP <- cor(HW3PR1[,1:4])
MP
```

```
##             Y         X1         X2         X3
## Y  1.00000000 0.24647538 0.06908391 0.81107297
## X1 0.24647538 1.00000000 0.06873854 0.06124793
## X2 0.06908391 0.06873854 1.00000000 0.12142333
## X3 0.81107297 0.06124793 0.12142333 1.00000000
```

```r
corrplot.mixed(MP, order='AOE')
```




```r
vif(q3_model)
```

```
##       X1       X2       X3
## 1.007623 1.018865 1.017869
```

3.1 (a)

Divide the 50 cases into two groups, placing the 25 cases with the smallest fitted values $\hat{Y}_i$ into group 1 and the other 25 cases into group 2. Conduct the Brown-Forsythe (BF) test for constancy of the error variance using $\alpha = 0.025$. State the hypotheses and your conclusion.

```r
# Conduct BF test by using anova() on

## Get fitted values, resids, for computations
HW3PR1$yhat <- fitted(q3_model)
HW3PR1$ei   <- residuals(q3_model)

# Sort by fitted values (ascending)
HW3_sorted <- HW3PR1[order(HW3PR1$yhat), ]

# Build the two groups
HW3_sorted$group <- rep(c(1,2), each = 25)

# get group medians
medians <- tapply(HW3_sorted$ei, HW3_sorted$group, median)

# Compute absolute deviations for each group (1,2)
HW3_sorted$di <- abs(HW3_sorted$ei - medians[HW3_sorted$group])

# Use ANOVA to run test for differences of variances
bf_model <- aov(di ~ factor(group), data = HW3_sorted)
summary(bf_model)
```

```
##               Df Sum Sq Mean Sq F value  Pr(>F)
## factor(group)  1  50875   50875   7.312 0.00945 **
## Residuals     48 333979    6958
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Brown-Forsythe Test is a test to evaluate whether the variances of independent groups are equal. We will use it here to tell whether our higher half of fitted values have error variance different from the lowest fitted values.
The assumptions required for the test are minimal, but it is still important that observations are independent. Additionally, since we are splitting the data into two groups, let us denote the error variance of the group of the smallest 25 fitted values as σ_1^2 and the error variance of the group of the largest 25 fitted values as σ_2^2.
The hypotheses for this test are:

$$H_0 : \text{The variance of errors in the two groups are the same } (\sigma_1^2 = \sigma_2^2)$$

$$H_1 : \text{The error variance in the two groups are not equivalent } (\sigma_1^2 \neq \sigma_2^2)$$

From our test, we attain a p-value of 0.00945, lower than our significance level of $\alpha = 0.025$. Therefore, we can reject the null hypothesis which stated that $\sigma_1^2 = \sigma_2^2$. Therefore, we have reason to believe that error variance changes across groups of observations grouped by fitted value. This means that we have evidence that heteroscedasticity is present across observations ordered by fitted value.

3.2 (b)

Obtain the externally Studentized residuals and identify any outlying Y observations. Use the Bonferroni outlier test procedure with $\alpha = 0.05$ (i.e., using $\alpha/50$ for each observation). State the decision rule and conclusion.

```r
# Externally Studentized residuals:
print(HW3b$ti)
```

```
##  [1] -0.24850979  1.18438006 -0.18765907 -0.47360892  0.63325556  0.17799082
##  [7] -1.09997822 -1.18262282 -0.98324208  2.05397311  0.94987777 -0.22574974
## [13] -0.41423047 -1.62056213  0.11760326 -0.94382745  1.29055075 -0.56267776
## [19]  0.74338731 -1.32921313 -0.30712634  0.22102471 -0.80552711  0.60752468
## [25]  0.37386810 -0.18628714  0.56093976 -0.38688071  0.45081240 -0.14243073
## [31] -0.98406109 -2.03633145  1.59391073  1.65797222 -1.64217518 -0.45504337
## [37] -0.99033238  2.10865604 -0.77054720  2.20066357  0.13423953  0.44061999
## [43]  0.79443071  0.20649779  0.12798372 -0.58475393 -1.16233998 -0.39703807
## [49] -0.02412907  1.61667652
```

```r
# Bonferroni outlier test:
HW3PR1 <- read.table("./HW3PR1.txt", header=TRUE)
q3_model <- lm(Y ~ X1 + X2 + X3, data = HW3PR1)
n <- nrow(HW3PR1)
p <- length(coef(q3_model)) #INCLUDES intercept
df <- n - p
alpha <- 0.05
```

```r
alpha_star <- alpha / n
t_crit <- qt(1 - alpha_star/2, df)
HW3PR1$ti <- rstudent(q3_model)
cat("\nBonferroni Critical Value:\n")
```

```
##
## Bonferroni Critical Value:
```

```r
print(t_crit)
```

```
## [1] 3.514957
```

```r
## p-value calculation
HW3PR1$pval <- 2 * pt(-abs(HW3PR1$ti), df)
# Identify outliers by cutoff
outlier_index <- which(abs(HW3PR1$ti) > t_crit)

cat("\nOutlier Detection Results:\n")
```

```
##
## Outlier Detection Results:
```

```r
if(length(outlier_index) == 0){
  cat("No Bonferroni outliers detected at alpha = 0.05.\n")
} else {
  cat("Bonferroni outliers detected at observations:\n")
  print(HW3PR1[outlier_index, ])
}
```

```
## No Bonferroni outliers detected at alpha = 0.05.
```

```r
# obs with p-value < alpha^star
bonf_p_outliers <- which(HW3PR1$pval < alpha_star)
if(length(bonf_p_outliers) > 0){
  cat("\nObservations with Bonferroni-adjusted p-value < alpha/n:\n")
  print(HW3PR1[bonf_p_outliers, c("case","ti","pval")])
}
```

We can start reviewing externally Studentized residuals for outliers by using the "rule of thumb" method, checking that the absolute value of these residuals are ≥ 2. Using this method, we find the observations 10, 32, and 38 to be outliers.
The Bonferroni outlier test procedures seeks to identify outliers more formally. In the Bonferroni outlier test procedure, we effectively split the significance level between observations, meaning that we must use a very strict threshold for identifying outliers. Using this method, we may only designate an observation as an outlier if its corresponding p-value $< \frac{0.05}{50} = 0.001$ for hypotheses:

$$H_0 : \text{Observation } i \text{ is not an outlier ,}$$

$$H_1 : \text{Observation } i \text{ is an outlier .}$$

The significance level shrinking is designed to counteract the chances of obtaining false positives since the method effectively uses n tests. Our decision rule can also be written as choosing to reject the null hypothesis when the absolute value of the test statistic exceeds a critical t-value, $t_{1-\frac{0.001}{2},\,45}$ (we have 45 degrees of freedom here because we have 50 observations, an intercept, and three predictors, and for the Bonferroni method we have df $n - p - 1 = 45$ degrees of freedom).
Using the Bonferroni method, we do not find any observations to be outliers (no null hypotheses rejected for our tests). It is somewhat unsurprising that we do not find any outliers using the Bonferroni method after having found outliers using an empirical method as the Bonferroni method is very conservative.

3.3 (c)

The cases are given in consecutive weeks. Prepare a time plot for of the residuals. Is there any indication that the error terms are correlated? Discuss.

```r
# Create dataframe w/ weeks, resids
resid_df <- data.frame(
  Week = 1:length(residuals(q3_model)),
  Residual = residuals(q3_model)
)
# build nice ggplot
ggplot(resid_df, aes(x = Week, y = Residual)) +
  geom_point(size = 3, color = "#408068") +
  geom_hline(yintercept = 0, linetype = "dashed",
             color = "black", linewidth = 0.8) +
  labs(title = "Time Series Plot of Residuals",
    x = "Week",
    y = "Residual") +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
```

```
    panel.grid.minor = element_blank())
```

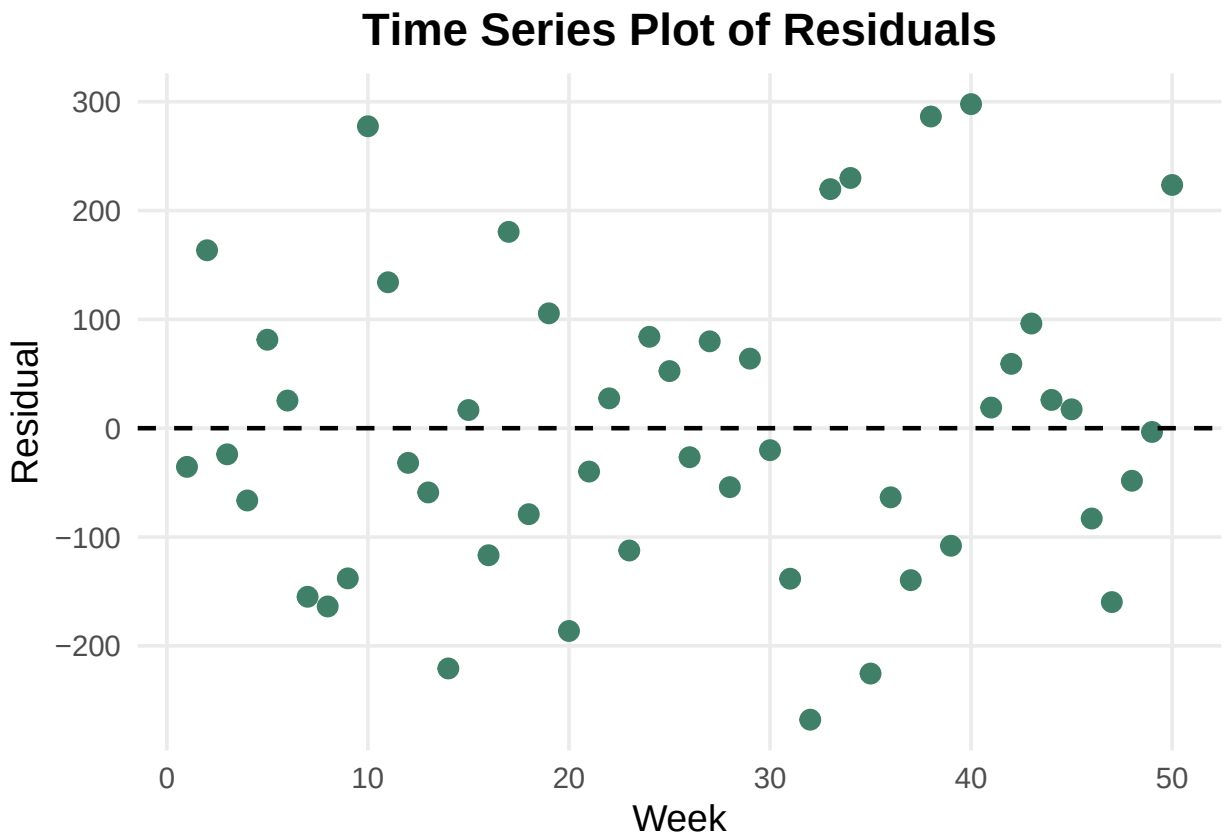


Figure 2: Time Series Plot of Residuals

Reviewing Figure 2, a time series plot of residuals in the model, we do not observe very clear seasonality in residual trends,. This does not mean that seasonality is not present, and indeed we may need data from several years to begin to clarify such a trend if it exists. In terms of seasonal trends, we may wonder if periods of higher variance in residuals (like the spring, and autumn) have anything to do with unexplained variance in the response that is particular to certain seasons.

3.4 (d)

The observations are listed in time order. Assume that the MLR model is appropriate. Test whether or not positive correlation is present; using $\alpha = 0.05$. State the hypotheses and conclusion.

```
dwtest(q3_model, alternative = "greater")
```

```
##
##  Durbin-Watson test
##
```

```
## data:  q3_model
## DW = 2.1727, p-value = 0.6786
## alternative hypothesis: true autocorrelation is greater than 0
```

Since the observations are ordered by time, we are interested in whether lag-1 autocorrelation exists in regression residuals. We will use a one-sided Durbin-Watson (DW) test to test for positive first-order autocorrelation of residuals to see if we must worry about this autocorrelation in our data analysis.
We will use hypotheses:

$$H_0 : \text{First-order autocorrelation } \rho = 0, \text{ and}$$

$$H_1 : \text{First-order autocorrelation } \rho > 0.$$

We are testing using an $\alpha = 0.05$ significance level.
We find a p-value of 0.6786, and therefore we do not have evidence from this test to suggest that there is positive autocorrelation between the residuals and their immediate neighbors.

3.5 (e)

Obtain the diagonal element of the hat matrix. Identify any outlying X observations using the rule of thumb presented in Note 6. Are any outlying Y observations also outlying X observations?

```r
# Get diagonal values
leverages <- hatvalues(q3_model)
# calculate n and p
n <- length(leverages)
p <- length(coef(q3_model))
# get cutoff value
cutoff <- 2 * p / n
# ID the high leverage observations using cutoff
high_leverage <- which(leverages >= cutoff)
# summary df
leverage_df <- data.frame(
  Observation = 1:n,
  Leverage = leverages,
  High_Leverage = leverages >= cutoff
)
# report results
cat("Cutoff value (2p/n):", round(cutoff, 3), "\n")
```

```
## Cutoff value (2p/n): 0.16
```

```r
cat("High leverage observations:", high_leverage, "\n")
```

```
## High leverage observations: 3 5 16 21 22 43 44 48
```

After pulling out diagonal elements of the hat matrices, the leverages, we can apply the rule of thumb in Note 6 which stated that we can identify an outlier in terms of leverage if we observe $h_{ii} \geq \frac{2p}{n}$. Here, we have $p = k + 1 = 4$ and $n = 50$. So, we should check for observations where $h_{ii} \geq 0.16$. These observatoins include observations $3, 5, 16, 21, 22, 43, 44, \$48$.
To check whether any outlying Y observations are also outlying X observations, we can recall the outlying Y observations from earlier: 10, 32, 38 (via Studentized residual cutoffs from Question 3b). These do not overlap with our set of outlying observations for X using this rule of thumb.

3.6 (f)

Obtain the DFFITs, DFBETAs, and Cook's distance values. Are any cases influential according to these measures? Discuss.

```r
## Sample size and number of parameters
n <- nobs(q3_model)
p <- length(coef(q3_model))

# DFFITS
dffits_vals <- dffits(q3_model)
## cutoff
dffits_cutoff <- 2 * sqrt(p / n)
influential_dffits <- which(abs(dffits_vals) > dffits_cutoff)
cat("DFFITS cutoff:", round(dffits_cutoff, 3), "\n")
```

```
## DFFITS cutoff: 0.566
```

```r
cat("Influential cases by DFFITS:", influential_dffits, "\n\n")
```

```
## Influential cases by DFFITS: 32
```

```r
# DFBETAS
dfbetas_vals <- dfbetas(q3_model)
## cutoff
dfbetas_cutoff <- 2 / sqrt(n)

# Identify influential cases (for any coefficient)
influential_dfbetas <- which(apply(abs(dfbetas_vals) > dfbetas_cutoff, 1, any))
cat("DFBETAS cutoff:", round(dfbetas_cutoff, 3), "\n")
```

```
## DFBETAS cutoff: 0.283
```

```r
cat("Influential cases by DFBETAS:", influential_dfbetas, "\n\n")
```

```
## Influential cases by DFBETAS: 10 14 16 32 43
```

```r
# Cook's Distance

cooks_vals <- cooks.distance(q3_model)
# curtoff "Rule of Thumb" Slide 63 Note 6
cooks_cutoff <- 0.8
influential_cooks <- which(cooks_vals > cooks_cutoff)
cat("Cook's Distance cutoff:", round(cooks_cutoff, 3), "\n")
```

```
## Cook's Distance cutoff: 0.8
```

```r
cat("Influential cases by Cook's Distance:", influential_cooks, "\n")
```

```
## Influential cases by Cook's Distance:
```

```r
cooks_vals <- cooks.distance(q3_model)
# curtoff another rule of thumb
cooks_cutoff <- 4/n
influential_cooks <- which(cooks_vals > cooks_cutoff)
cat("Cook's Distance cutoff:", round(cooks_cutoff, 3), "\n")
```

```
## Cook's Distance cutoff: 0.08
```

```r
cat("Influential cases by Cook's Distance:", influential_cooks, "\n")
```

```
## Influential cases by Cook's Distance: 32
```

DFFITs:
According to this measure, the following case is influential: 32
DFBETAs:
According to this measure, the following cases are influential: 10, 14, 16, 32, 43
Cook's distance:
According to this measure, no cases are influential using the universal rule of thumb of a 0.8 cutoff value. However, using a $4/n = 0.08$ cutoff suggests that observation 32 is influential.
Discussion:
We try out several methods for high-influence observation detection, and find that observation 32 repeatedly shows up, so we may question how the influence of that point is affected our fitted model. We see that the DFFITS method and Cook's distance method (with $4/n$ cutoff in use) give identical results for this particular set of data, both identifying only observation 32 as influential. We find the DFBETAs method to be more liberal in its selection of influential points, and we observe that using its rule-of-thumb cutoff for large samples, we observe 5 influential observations.

3.7 (g)

Obtain the scatter plot matrix and the correlation matrix. What do these matrices show about pairwise linear associations among the predictors?

```
pairs(HW3PR1[, c("Y", "X1", "X2", "X3")], pch = 20)
```

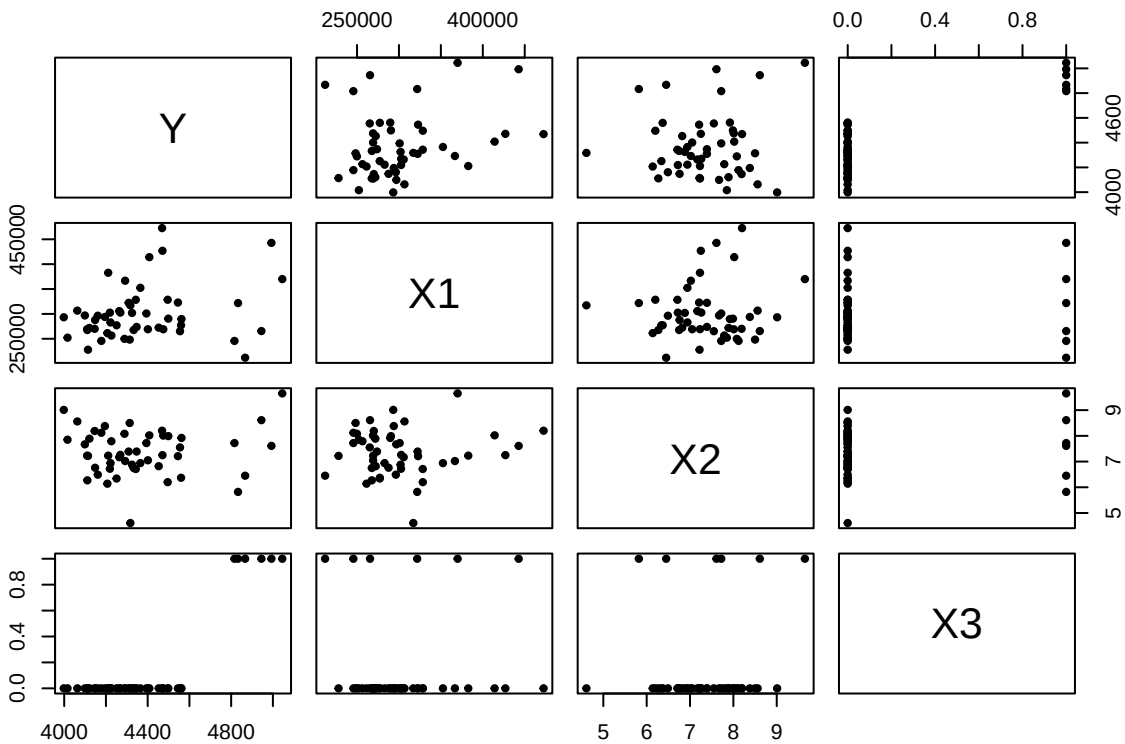


Figure 3: Scatter Plot Matrix - Q3 Response and Predictors

```
corrplot.mixed(MP, order='AOE')
```

Figure 3 shows the scatterplot matrix of the predictors and response for this problem. We observe some important things about the pairwise relationships of our predictors and response. Perhaps the most notable plot is the relationship between $X3$ and Y. In that pair's plots, we observe that $X3$ is a binary variable, but that we have a perfect grouping of high Y values having $X3 = 1$ and relatively lower Y values having $X3 = 0$. This relationship should prove very useful for regression. For the other plots, we obtain point clouds that do not very clearly seem to indicate strong collinearity. Even the value of $X3$ seems to be somewhat scattered among values of $X1$ and $X2$.
Figure 4 helps us quantify the relationships we view with the scatterplot matrix. In the scatterplot matrix, we do not find the relationships between predictors to have very high absolute correlation. The only pairing with high correlation is $X3$ and Y, which is okay in terms of collinearity because it is not a predictor-predictor relationship. The correlations of X_i's are all $\leq |0.12|$, which is likely largely inconsequential for collinearity diagnostics.

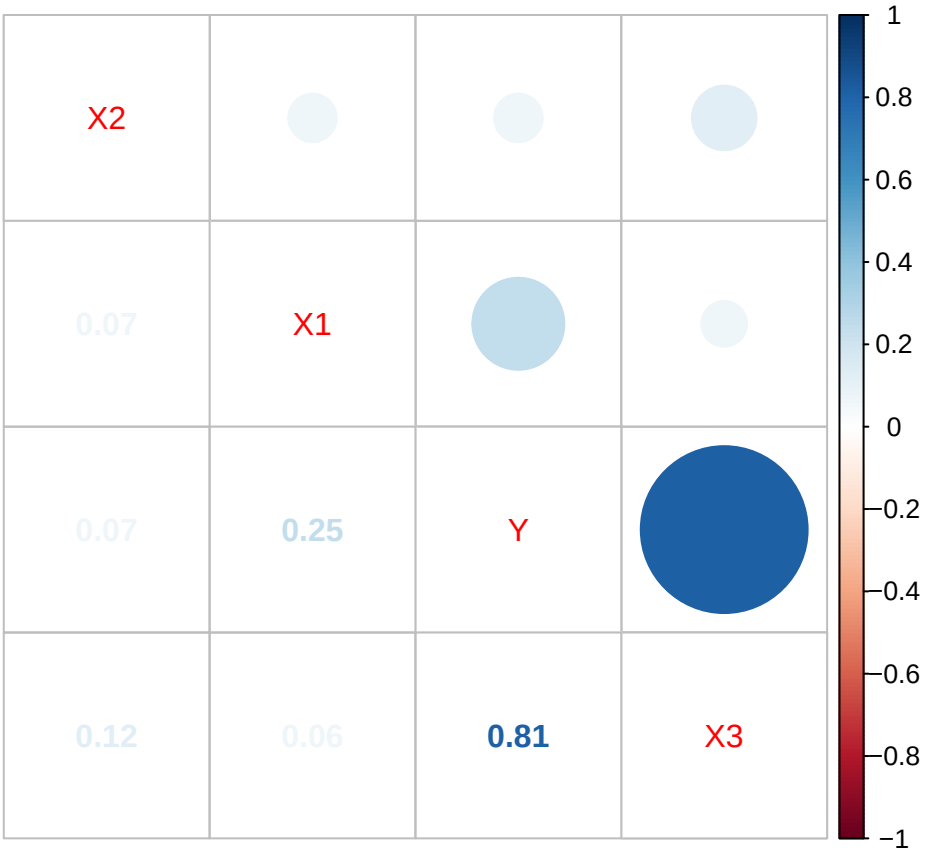


Figure 4: Correlation Matrix - Q3 Response and Predictors

3.8 (h)

Find the three variance inflation factors. Do they indicate that a serious multicollinearity problem exists here?

```r
vif(q3_model)
```

```
##       X1       X2       X3
## 1.007623 1.018865 1.017869
```

We get variance inflation factors very close to 1 (the minimum for VIF), suggesting that we do *not* observe a serious multicollinearity problem here. In fact, the multicollinearity is extremely minor here.

4 Problem 4

For the data $(X_1, X_2, Y) = (-1, -1, 7.2), (-1, 0, 8.1), (0, 0, 9.8)$. Suppose that we fit a MLR model with an intercept to this data set. In this case, $n = 3$, $k = 2$, and $p = 3$. Compute the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. What are h_{ii}, $\hat{Y}_i$, and e_i for $i = 1, 2, 3$?

```r
# Dataframe for data
q4_df <- data.frame(
  X1 = c(-1, -1, 0),
  X2 = c(-1,  0, 0),
  Y  = c(7.2, 8.1, 9.8)
)

# MLR model fit
q4_mlr_model <- lm(Y ~ X1 + X2, data = q4_df)

# build the design matrix X
X <- model.matrix(q4_mlr_model)

# find hat matrix
H <- X %*% solve(t(X) %*% X) %*% t(X)

# print hat matrix
H
```

```
##   1            2            3
## 1 1 1.110223e-16 1.110223e-16
## 2 0 1.000000e+00 1.110223e-16
## 3 0 0.000000e+00 1.000000e+00
```

```r
# influence matrix
h_ii <- diag(H)

# fitted values
yhat <- fitted(q4_mlr_model)
resid <- residuals(q4_mlr_model)

# give each observation its own df
obs_1 <- data.frame(h_ii = h_ii[1], yhat = yhat[1], resid = resid[1])
obs_2 <- data.frame(h_ii = h_ii[2], yhat = yhat[2], resid = resid[2])
obs_3 <- data.frame(h_ii = h_ii[3], yhat = yhat[3], resid = resid[3])

# Report
cat("Observation 1 influence:", obs_1$h_ii,
    "; fitted value:", obs_1$yhat,
    "; residual:", obs_1$resid, "\n")
```

```
## Observation 1 influence: 1 ; fitted value: 7.2 ; residual: 0
```

```r
cat("Observation 2 influence:", obs_2$h_ii,
    "; fitted value:", obs_2$yhat,
    "; residual:", obs_2$resid, "\n")
```

```
## Observation 2 influence: 1 ; fitted value: 8.1 ; residual: 0
```

```r
cat("Observation 3 influence:", obs_3$h_ii,
    "; fitted value:", obs_3$yhat,
    "; residual:", obs_3$resid, "\n")
```

```
## Observation 3 influence: 1 ; fitted value: 9.8 ; residual: 0
```

We are fitting an MLR model with coefficients $\beta_0, \beta_1, \beta_2$ to a dataset with only three points. Therefore, our regression plane will pass through each of the points because a plane can always pass through three points in three-dimensional space. We find all observations to have an influence of 1, and fitted values equal to their Y value (leaving all residuals as 0).

5 Problem 5

Increased arterial blood pressure in the lungs frequently leads to the development of heart failure in patients with chronic obstructive pulmonary disease (COPD). The standard method for determining arterial lung pressure is invasive, technically difficult, and involves some risk to the patient. The radionuclide imaging technique is a noninvasive, less risky method for estimating arterial pressure in the lungs. To investigate the predictive ability of this method, a cardiologist collected data on 19 mild to moderate COPD patients. In the data set, the response variable (Y) is the invasive measure of systolic pulmonary arterial pressure, and the potential noninvasive predictors are the emptying rate of blood into the pumping chamber of the heart (X_1) and the ejection rate of blood pumped out of the heart into the lungs (X_2).

Consider the following model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

where $\epsilon \sim N(0, \sigma^2)$. The resulting ANOVA table is given below.

Table 1: Incomplete ANOVA Table

Analysis of Variance					
Source	DF	Sum of Square	Mean Square	F Stat	Prob > F
Model	*	4961.46426	*	*	*
Error	*	*	*		
C Total	*	8087.68421			

5.1 (a)

Fill in the missing values in Table 1.

```r
# p-value calculation:
## Define test F statistic
q5a_test_F <- 12.69639

## Calculate p-value
q5a_pval <- 1 - pf(q5a_test_F, 2, 16)
cat("p-value: ", q5a_pval)
```

```
## p-value:  0.0004983829
```

Although we begin with very few pieces of information, they will prove sufficient to complete our ANOVA table. Besides the existing table's information, we will also use the following information from the problem description: $n = 19$, $k = 2$ (we have two predictors), $p = 3$ (predictors + intercept). Let's find each of our eight missing values:

Model DF:
The degrees of freedom associated with the model are $k = 2$.

Error DF:
The error degrees of freedom are $n - p = 19 - 3 = 16$.

C Total DF:
The corrected total degrees of freedom is the sum of the prior two degrees of freedom, here 18.

Error Sum of Squares:
Since the sum of squares are additive, we can subtract the regression sum of squares from the total sum of squares to yield SSE $= 8087.68421 - 4961.46426 = 3126.21995$.

Model Mean Square:
The regression MS is $\text{MS}_{\text{MODEL}} = \frac{\text{SSR}}{\text{df}_{\text{MODEL}}} = \frac{4961.46426}{2} = 2480.73213$

Error Mean Square:
The MSE is $\text{MSE} = \frac{\text{SSE}}{\text{df}_{\text{ERROR}}} = \frac{3126.21995}{16} = 195.38875$

F Statistic:
The F statistic for the regression model can be calculated as $\frac{MS_{\text{MODEL}}}{MSE} = \frac{2480.73213}{195.38875} = 12.69639$

p-value accompanting F-statistic:
Finally, we can calculate a p-value to accompany the aforementioned F statistic. This p-value will be calculated knowing that the test statistic follows an $F_{2,16}$ distribution. We can find the probability that an $F_{2,16}$ random variable is greater than our observed test statistic, 12.69639, and find the p-value as p-value $= 1 - \Pr(F_{2,16} > F_{\text{test}})$. We will use R to calculate the explicit value which works out to be 0.0004983829.

Table 2: Completed ANOVA Table

Analysis of Variance					
Source	DF	Sum of Square	Mean Square	F Stat	Prob > F
Model	2	4961.46426	2480.73213	12.69639	0.0004983829
Error	16	3126.21995	195.38875		
C Total	18	8087.68421			

5.2 (b)

The estimates of the model parameters are given in the following table.

Table 2: Parameter Estimates

Variable	DF	Estimate	Std Error	T Stat	Prob > |T|	VIF
Intercept	1	84.215024	10.25104824	8.215	0.0001	0
X_1	1	-0.587215	*	*	0.1522	*
X_2	1	-0.494677	0.18584188	-2.662	0.0171	*

5.2.1 (i)

What are the "Std Error" and "T Stat" for X_1?

```r
# Calculate test statistic using distribution t_1 and p-value:
q5_b_i_test_stat <- qt(1 - 0.1522/2, df = 16)
cat("test statistic absolute value:", q5_b_i_test_stat)
```

```
## test statistic absolute value: 1.503466
```

Standard Error for X_1:
We can find the standard error for X_1 by recognizing that the test statistic can be formulated as a ratio of an estimate of the slope for X_1 and its standard error. This leads us to recognize that $\text{SE}(\hat{\beta}_1) = \frac{\hat{\beta}_1}{t}$. Unfortunately we are also missing our test statistic to we have to keep digging. The magnitude of the test statistic can be retrieved from our p-value when we recognize that the p-value for a two sided test here is p-value $= 2\Pr(T_{df} > |t|)$. We are given a p-value of 0.1522 for X_1's coefficient here. We can also recognize that 16 degrees of freedom are associated with the term (error df). So, we are seeking the test statistic such that a random variable $T \sim t_{16}$ gives our p-value:

$$0.1522 = 2\Pr(T_{16} > t_{\text{test}}) \implies |t| = t_{1-\frac{p}{2},\,16} = 1.503.$$

Now, we want to ensure that our test statistic has the proper sign. Since our regression coefficient was negative, it should be negative. Therefore, our test statistic for X_1 should be -1.503.
Finally, we can proceed to our calculation of the standard error for X_1's regression coefficient:

$$\text{SE}(\hat{\beta}_1) = \frac{\hat{\beta}_1}{t} = \frac{-0.587215}{-1.503} = 0.391$$

T statistic for X_1:
The T statistic for X_1 was identified to find the standard error above. We found it to be -1.503.

5.2.2 (ii)

It is calculated that the Pearson correlation coefficient between X_1 and X_2 is $r = 0.65285$. Obtain VIF_1 and VIF_2 for X_1 and X_2, respectively. Is there any evidence indicating that a serious multicollinearity problem exists here?

Since we do not have the full data here, we cannot simply use vif() to get our variance inflation factors. Instead we will manually compute them:
With only two predictors in our MLR model, the VIFs will actually be symmetric because we formulate the VIF as $\text{VIF}_i = \frac{1}{1-R_i^2}$. But, here $R_1^2 = r^2$ and $R_2^2 = r^2$ so:

$$\text{VIF}_1 = \frac{1}{1-R_1^2} = \frac{1}{1-0.65285^2} = 1.7428, \text{ and}$$

$$\text{VIF}_2 = \frac{1}{1-R_2^2} = 1.7428.$$

Our VIF values are quite low, so serious collinearity is not a concern for this modeling endeavor.

5.2.3 (iii)

What is the p-value for testing the null hypothesis H_0: $\beta_2 = 0$ versus one-sided alternative hypothesis H_1: $\beta_2 > 0$?

```
# Calculate p-value:
```

Given the test statistic for X_2 of $t = -2.662$ and the two-sided p-value of 0.0171, we want to find a one sided test statistic for the scenario in which we believe that β_2 may be greater than zero.
Before even starting to calculate a particular one-sided p-value for this scenario we should consider that we have a quite strong-magnitude *negative* test statistic -2.662, implying that we will get a very high one-sided p-value for our new test.
Anyway, we can relate the one-sided p-value to the two-sided p-value via the formula $\text{p-value}_{\text{one-sided}} = 1 - \frac{\text{p-value}_{\text{two-sided}}}{2} = 1 - \frac{0.0171}{2} = 0.99145$. As we expected, the p-value for this test is very high.